

Proof of Conjecture 00000000035: monochromatic Schur triples with $xy + 1$ prime, $N = 17$

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Abstract

Conjecture 00000000035 asserts that there is an N such that every 2-coloring of $[N] = \{1, \dots, N\}$ contains a monochromatic Schur triple (x, y, z) , $x + y = z$, with $xy + 1$ prime. We prove this with $N = 17$, even when $x < y$ is required. The value 17 is optimal for $x < y$: an explicit coloring of [16] avoids all such triples. (If $x = y$ is allowed, $N = 7$ already suffices.) The proof is a finite case analysis, fully checked in Lean 4 (core library only) and by exhaustive search in Python.

1 Statement and reading

The conjecture (English) reads:

There exists $N(k)$ such that any 2-coloring of $[N]$ contains a monochromatic Schur triple (x, y, z) with $x + y = z$ and $xy + 1$ prime.

The Chinese version is identical. A Schur triple in $[N]$ is $x, y, z \in [N]$ with $x + y = z$. The parameter k in “ $N(k)$ ” does not occur anywhere else in the statement, and the coloring is explicitly a 2-coloring. So the statement is: *there exists N such that every $c : [N] \rightarrow \{R, B\}$ has $x, y, z \in [N]$ with $x + y = z$, $xy + 1$ prime and $c(x) = c(y) = c(z)$* . Since $N = 17$ works for every value of k , any dependence on k is harmless. We prove the stronger version that requires $x < y$, which in particular makes x, y, z distinct.

2 The triples

Let $\mathcal{T}$ be the set of (x, y, z) with $1 \leq x < y$, $z = x + y \leq 17$ and $xy + 1$ prime. There are exactly 31 of them:

(1, 2, 3), (1, 4, 5), (1, 6, 7), (1, 10, 11), (1, 12, 13), (1, 16, 17), (2, 3, 5), (2, 5, 7), (2, 6, 8),
(2, 8, 10), (2, 9, 11), (2, 11, 13), (2, 14, 16), (2, 15, 17), (3, 4, 7), (3, 6, 9), (3, 10, 13),
(3, 12, 15), (3, 14, 17), (4, 7, 11), (4, 9, 13), (4, 10, 14), (4, 13, 17), (5, 6, 11), (5, 8, 13),
(5, 12, 17), (6, 7, 13), (6, 10, 16), (6, 11, 17), (7, 10, 17), (8, 9, 17).

For example $1 \cdot 2 + 1 = 3$, $8 \cdot 9 + 1 = 73$ and $7 \cdot 10 + 1 = 71$ are prime. Only the *presence* of these triples is used; that each $xy + 1$ is prime is checked in Lean with a verified trial-division checker.

3 Proof that $N = 17$ works

Theorem 1. *Every 2-coloring c of [17] has a monochromatic triple in $\mathcal{T}$.*

Proof. Suppose c has none. Then for each $(x, y, z) \in \mathcal{T}$ the three colors are not all equal. Hence, *if two of them are equal, the third has the other color*. Write “ $(x, y, z) \Rightarrow c(w) = \cdot$ ” for this deduction. We branch on a few values (a branching value is written without an arrow) and propagate. Every branch ends with a triple of $\mathcal{T}$ whose three colors are equal, which is a contradiction. The 12 branches below cover all cases: at each bold split both colors are taken (all 12 combinations that arise are listed).

1. $c(2) = R, c(17) = R, (2, 15, 17) \Rightarrow c(15) = B, c(3) = R, (1, 2, 3) \Rightarrow c(1) = B, (2, 3, 5) \Rightarrow c(5) = B, (1, 4, 5) \Rightarrow c(4) = R, (3, 4, 7) \Rightarrow c(7) = B, (1, 6, 7) \Rightarrow c(6) = R, (2, 6, 8) \Rightarrow c(8) = B, (3, 6, 9) \Rightarrow c(9) = B, (3, 14, 17) \Rightarrow c(14) = B, (4, 13, 17) \Rightarrow c(13) = B, (5, 8, 13)$ monochromatic.
2. $c(2) = R, c(17) = R, (2, 15, 17) \Rightarrow c(15) = B, c(3) = B, (3, 12, 15) \Rightarrow c(12) = R, (5, 12, 17) \Rightarrow c(5) = B, c(6) = R, (2, 6, 8) \Rightarrow c(8) = B, (5, 8, 13) \Rightarrow c(13) = R, (1, 12, 13) \Rightarrow c(1) = B, (1, 4, 5) \Rightarrow c(4) = R, (4, 13, 17)$ monochromatic.
3. $c(2) = R, c(17) = R, (2, 15, 17) \Rightarrow c(15) = B, c(3) = B, (3, 12, 15) \Rightarrow c(12) = R, (5, 12, 17) \Rightarrow c(5) = B, c(6) = B, (3, 6, 9) \Rightarrow c(9) = R, (2, 9, 11) \Rightarrow c(11) = B, (5, 6, 11)$ monochromatic.
4. $c(2) = R, c(17) = B, c(3) = R, (1, 2, 3) \Rightarrow c(1) = B, (1, 16, 17) \Rightarrow c(16) = R, (2, 3, 5) \Rightarrow c(5) = B, (1, 4, 5) \Rightarrow c(4) = R, (2, 14, 16) \Rightarrow c(14) = B, (3, 4, 7) \Rightarrow c(7) = B, (1, 6, 7) \Rightarrow c(6) = R, (2, 6, 8) \Rightarrow c(8) = B, (3, 6, 9) \Rightarrow c(9) = B, (8, 9, 17)$ monochromatic.
5. $c(2) = R, c(17) = B, c(3) = B, (3, 14, 17) \Rightarrow c(14) = R, (2, 14, 16) \Rightarrow c(16) = B, (1, 16, 17) \Rightarrow c(1) = R, c(6) = R, (1, 6, 7) \Rightarrow c(7) = B, (2, 6, 8) \Rightarrow c(8) = B, (3, 4, 7) \Rightarrow c(4) = R, (1, 4, 5) \Rightarrow c(5) = B, (4, 10, 14) \Rightarrow c(10) = B, (7, 10, 17)$ monochromatic.
6. $c(2) = R, c(17) = B, c(3) = B, (3, 14, 17) \Rightarrow c(14) = R, (2, 14, 16) \Rightarrow c(16) = B, (1, 16, 17) \Rightarrow c(1) = R, c(6) = B, (3, 6, 9) \Rightarrow c(9) = R, (2, 9, 11) \Rightarrow c(11) = B, (6, 11, 17)$ monochromatic.
7. $c(2) = B, c(17) = R, c(3) = R, (3, 14, 17) \Rightarrow c(14) = B, (2, 14, 16) \Rightarrow c(16) = R, (1, 16, 17) \Rightarrow c(1) = B, c(6) = R, (3, 6, 9) \Rightarrow c(9) = B, (2, 9, 11) \Rightarrow c(11) = R, (6, 11, 17)$ monochromatic.
8. $c(2) = B, c(17) = R, c(3) = R, (3, 14, 17) \Rightarrow c(14) = B, (2, 14, 16) \Rightarrow c(16) = R, (1, 16, 17) \Rightarrow c(1) = B, c(6) = B, (1, 6, 7) \Rightarrow c(7) = R, (2, 6, 8) \Rightarrow c(8) = R, (3, 4, 7) \Rightarrow c(4) = B, (1, 4, 5) \Rightarrow c(5) = R, (4, 10, 14) \Rightarrow c(10) = R, (7, 10, 17)$ monochromatic.
9. $c(2) = B, c(17) = R, c(3) = B, (1, 2, 3) \Rightarrow c(1) = R, (1, 16, 17) \Rightarrow c(16) = B, (2, 3, 5) \Rightarrow c(5) = R, (1, 4, 5) \Rightarrow c(4) = B, (2, 14, 16) \Rightarrow c(14) = R, (3, 4, 7) \Rightarrow c(7) = R, (1, 6, 7) \Rightarrow c(6) = B, (2, 6, 8) \Rightarrow c(8) = R, (3, 6, 9) \Rightarrow c(9) = R, (8, 9, 17)$ monochromatic.
10. $c(2) = B, c(17) = B, (2, 15, 17) \Rightarrow c(15) = R, c(3) = R, (3, 12, 15) \Rightarrow c(12) = B, (5, 12, 17) \Rightarrow c(5) = R, c(6) = R, (3, 6, 9) \Rightarrow c(9) = B, (2, 9, 11) \Rightarrow c(11) = R, (5, 6, 11)$ monochromatic.
11. $c(2) = B, c(17) = B, (2, 15, 17) \Rightarrow c(15) = R, c(3) = R, (3, 12, 15) \Rightarrow c(12) = B, (5, 12, 17) \Rightarrow c(5) = R, c(6) = B, (2, 6, 8) \Rightarrow c(8) = R, (5, 8, 13) \Rightarrow c(13) = B, (1, 12, 13) \Rightarrow c(1) = R, (1, 4, 5) \Rightarrow c(4) = B, (4, 13, 17)$ monochromatic.
12. $c(2) = B, c(17) = B, (2, 15, 17) \Rightarrow c(15) = R, c(3) = B, (1, 2, 3) \Rightarrow c(1) = R, (2, 3, 5) \Rightarrow c(5) = R, (1, 4, 5) \Rightarrow c(4) = B, (3, 4, 7) \Rightarrow c(7) = R, (1, 6, 7) \Rightarrow c(6) = B, (2, 6, 8) \Rightarrow c(8) = R, (3, 6, 9) \Rightarrow c(9) = R, (3, 14, 17) \Rightarrow c(14) = R, (4, 13, 17) \Rightarrow c(13) = R, (5, 8, 13)$ monochromatic.

Each branch is a complete chain of forced deductions from the listed triples, so no coloring avoids $\mathcal{T}$. $\square$

Theorem 2. *For every k there is $N(k)$ (namely 17) such that every 2-coloring of $[N]$ contains a monochromatic Schur triple (x, y, z) , $x + y = z$, with $xy + 1$ prime. Conjecture 00000000035 is true.*

4 Sharpness

Theorem 3. *The coloring of $1, 2, \dots, 16$ given by 0010101110110101 ($0 = R$, $1 = B$) has no monochromatic triple $(x, y, x + y)$ with $1 \leq x < y$, $x + y \leq 16$ and $xy + 1$ prime. So 17 is the least admissible N when $x < y$ is required.*

Proof. There are 23 such triples in [16], and each is checked directly (Lean `good16_check`, Python `verify.py`). $\square$

Remark 4. If $x = y$ is allowed, the extra triples $(x, x, 2x)$ with $x^2 + 1$ prime appear ($x = 1, 2, 4, 6, \dots$), and the least N drops to 7 (`verify.py`). The conjecture only asks for existence, so either reading is proved by $N = 17$.

5 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib).

- `Prime n` is the standard definition ($n \geq 2$ and every divisor is 1 or n). `prime_of_check` and `check_of_prime` prove that a Boolean trial-division checker decides it.
- `SchurPrimeTriple N x y z` means $1 \leq x < y$, $x + y = z \leq N$ and `Prime (x*y+1)`. Colorings are functions $c : \mathbb{N} \rightarrow \text{Bool}$; only the values on $[N]$ matter.
- `schur_17`: every c has a monochromatic `SchurPrimeTriple 17`. The proof is exactly the case analysis above, generated mechanically: 11 case splits, and each propagation step uses one triple via the lemmas `force_x`, `force_y`, `force_z` and `clash`.
- `conjecture_00000000035`: $\forall k, \exists N, \forall c, \exists x y z, \text{SchurPrimeTriple } N \ x \ y \ z \wedge c(x) = c(y) = c(z)$. `conjecture_00000000035_le` is the variant that only requires $x \leq y$.
- `sixteen_avoids`: the coloring `good16` of [16] avoids all triples.

`lake build` succeeds. There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` shows at most `propxt`, `Classical.choice` and `Quot.sound`.

6 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # brute force over all 2^17 colorings
pdflatex report.tex
```